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Smart Calorie and Nutrition Calculator Using C++

Team 5



Problem

Maintaining adequate calorie and nutrient intake is important for supporting healthy daily habits. However, many people may have limited knowledge of their daily nutritional requirements and find manual calculations difficult.



Low Health Literacy

Lack of basic knowledge about their body's daily calorie and nutrition needs.

Health Consequences

Poor eating habits leading to malnutrition, obesity, and preventable chronic health issues.

Calculation Barrier

Difficulty in calculating precise daily targets manually

So, the solution?

Smart Calorie and Nutrition Calculator Using C++

- Replaces confusing health formulas with instant and personalized calculations.
- Delivers precise daily calorie and macronutrient goals to keep your body healthy.

Elbel, B., et al. (2013). "Consumers' estimation of calorie content at fast food restaurants." *BMJ*, 346, f2811.

Basic algorithm and system

How it works???

- **User Input**

1. Gender
2. W = Weight (in kg)
3. H = Height (in cm)
4. A = Age (in years)
5. Daily Activity Factor

- **Daily Activity Factor:**

1. Little to no exercise: 1.2
2. Light Activity (1-3 days/week): 1.375
3. Moderate Activity (3-5 days/week): 1.55
4. Heavy Activity (6-7 days/week): 1.725
5. Very Heavy Activity / Athlete: 1.9

- **BMR Calculation (Basal Metabolic Rate)**

: Calories your body burns at complete rest just to stay alive.

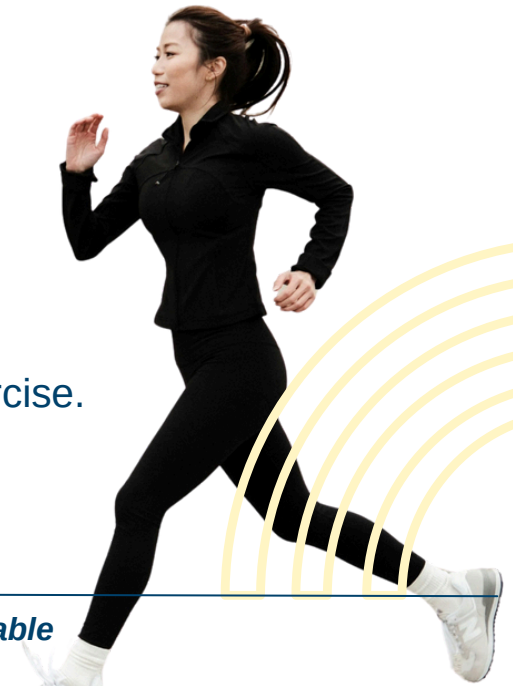
Men: $BMR = 10W + 6.25 H - 5A + 5$

Women: $BMR = 10W + 6.25 H - 5A - 161$

- **TDEE Calculation (Total Daily Energy Expenditure)**

: Total calories your body burns in a full day, including all movement and exercise.

$TDEE = BMR \times \text{Activity Factor}$





• User Input Validation

Data type:

Gender (integer) :

1 (male)

2 (female)

Weight and Height (double)

Age (integer)

Activity option (integer)

Activity factor (double)

The program asks the user to enter their gender, weight, height, and age. Each input is validated to make sure that the data is valid. If not, user got error message and have to re-input the value until its valid

• Daily Activity Factor

The user then selects their daily activity level from 1 to 5, and the program assigns the corresponding activity factor using a **switch** statement.

• BMR and TDEE

The program calculates the user's BMR based on their gender, weight, height, and age. The BMR is then multiplied by the selected activity factor to calculate the TDEE.

Macronutrient Target Setting

- **User Input**

1. Target Protein (Range 10%-35%)
2. Target Carbs (Range 45%-65%)
3. Target Fat (Range 20%-35%)

- The program uses a **do-while** loop to input the target percentages and check whether each input is within the required range. If the input is out of range, the program will show an error message and ask the user to re-enter the percentages until the input is valid.
- The program also initializes the variable `remainingPersen` to 100. The value of `remainingPersen` is then reduced by the inputted target protein and carbohydrate percentages.
- After calculating the protein and carbohydrate percentages, the remaining percentage is assigned to fat. Therefore, the total percentage of protein, carbohydrates, and fat will always be 100%. The program then checks whether the calculated fat percentage is within the required range of 20%–35%. If it is outside this range, the program will display an error message and ask the user to re-enter the target percentages.

• Calculate to grams:

$$1. \text{Protein (g)} = \frac{\text{TDEE} \times \% \text{Target Protein}}{4}$$

$$2. \text{Carbs (g)} = \frac{\text{TDEE} \times \% \text{Target Carbs}}{4}$$

$$3. \text{Fat (g)} = \frac{\text{TDEE} \times \% \text{Target Fat}}{9}$$



Example

Sample Input Data:

1. Gender: Male
2. Weight (W): 70 kg
3. Height (H): 175 cm
4. Age (A): 25 years
5. Activity Level: Moderate Activity (1.55)
6. Macro Target Split: 20% Protein, 55% Carbohydrates, 25% Fat

Step-by-step

$$\text{BMR} = 10W + 6.25 H - 5A + 5$$

$$= 10(70) + 6.25(175) - 5(25) + 5$$

$$= 1673.75 \text{ kcal}$$

$$\text{TDEE} = \text{BMR} \times \text{Activity Factor}$$

$$= 1673.75 \times 1.55 = 2594.31 \text{ kcal}$$

Protein (20% Target)

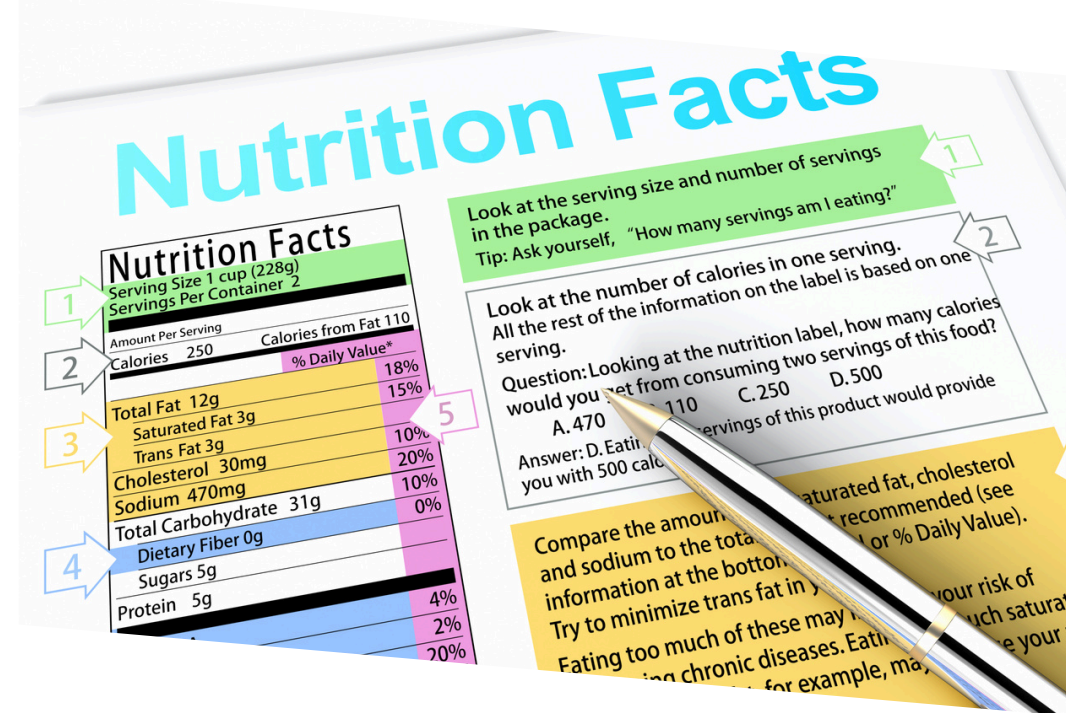
$$= \frac{2594.31 \times 0.20}{4} = 129.7 \text{ g}$$

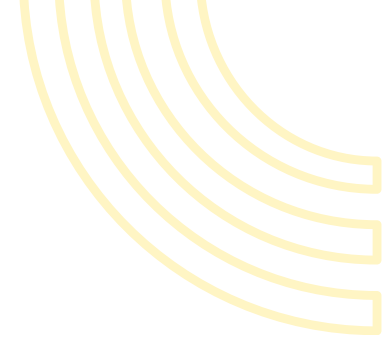
Carbohydrates (55% target)

$$= \frac{2594.31 \times 0.55}{4} = 356.7 \text{ g}$$

Fat (25% target)

$$= \frac{2594.31 \times 0.25}{9} = 72.1 \text{ g}$$



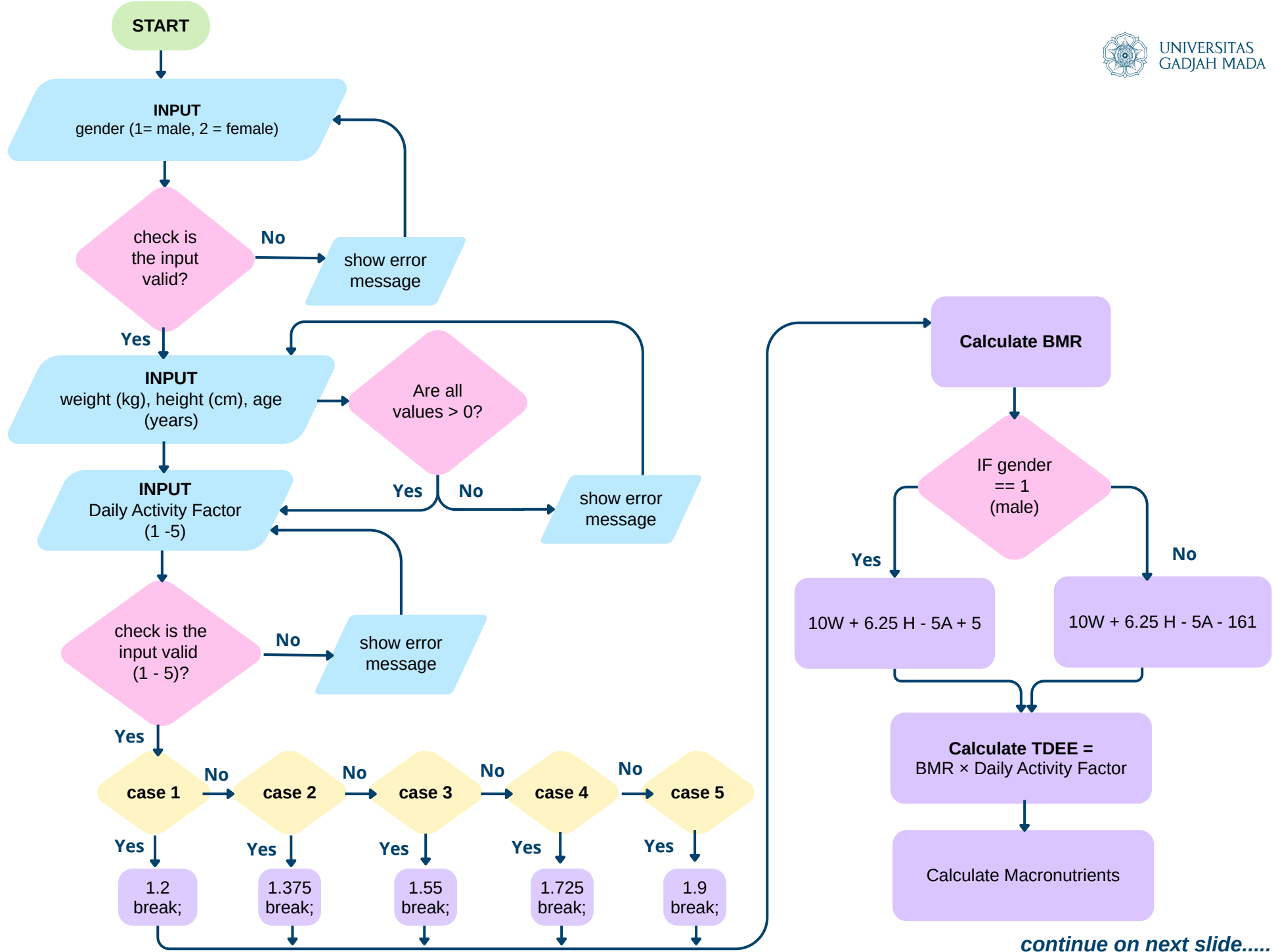


Example Output

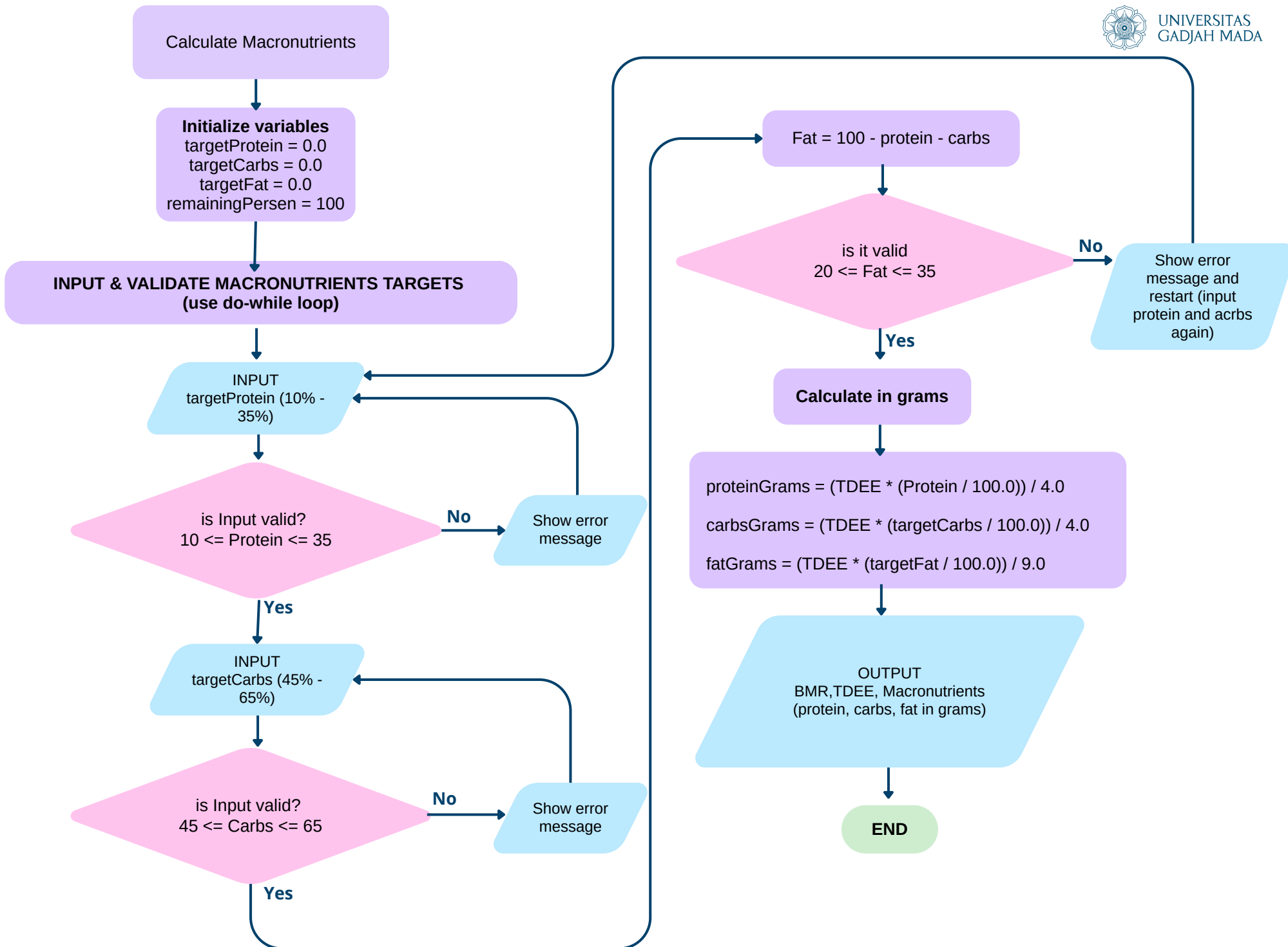
Parameter	Value
BMR	1.674 kcal/day
TDEE	2.594 kcal/day
Protein (20%)	130 g/day
Carbs (55%)	357 g/day
Fat (25%)	72 g/day

To keep things simple, C++ uses fixed << setprecision(0) to turn this into non decimal numbers

Flowchart Detail



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Conclusion

This C++ program will:

- Shows you exactly what your body needs every day in clear and simple numbers.
- Helps you build healthy habits to avoid problems like obesity and malnutrition.
- Handles all the complicated formulas instantly so you don't have to do it manually.



References

Black, A. E. (1996). Physical activity levels in adults: Practical implications for estimating energy requirements. *European Journal of Clinical Nutrition*, 50(2), 101-113.

Elbel, B., et al. (2013). "Consumers' estimation of calorie content at fast food restaurants." *BMJ*, 346, f2811.

Institute of Medicine. (2005). *Dietary Reference Intakes for energy, carbohydrate, fiber, fat, fatty acids, cholesterol, protein, and amino acids*. The National Academies Press.

Mifflin, M. D., St Jeor, S. T., Hill, L. A., Scott, B. J., Daugherty, S. A., & Koh, Y. O. (1990). A new predictive equation for resting energy expenditure in healthy individuals. *The American Journal of Clinical Nutrition*, 51(2), 241-247.



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Thank You.



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